

Product ID	Model number	Bus power [W] ¹⁾	Internal I/O power [W]
DC1396	X20DC1396	-0.01	-1.4
DC1398	X20DC1398	-0.01	-1.3
DC2395	X20DC2395	-0.01	-1.4
DC2396	X20DC2396	-0.01	-1.5
DC2398	X20DC2398	-0.01	-1.4
DC4395	X20DC4395	-0.01	-1.5
DI2371	X20DI2371	-0.12	-0.26
DI2377	X20DI2377	-0.2	-0.74
DI4371	X20DI4371	-0.15	-0.52
DI6371	X20DI6371	-0.16	-0.78
DI6372	X20DI6372	-0.16	-0.78
DI9371	X20DI9371	-0.2	-
DI9372	X20DI9372	-0.2	-1.56
DO2322	X20DO2322	-0.14	-0.2
DO4322	X20DO4322	-0.15	-0.31
DO4332	X20DO4332	-0.15	-0.31
DO6321	X20DO6321	-0.16	-0.5
DO6322	X20DO6322	-0.16	-0.5
DO8332	X20DO8332	-0.21	-
DO9321	X20DO9321	-0.21	-0.99
DO9322	X20DO9322	-0.21	-0.99
PS2100	X20PS2100	-0.2	+240.0 ²⁾
PS3300	X20PS3300	+7.0	+240.0 ²⁾
PS9400	X20PS9400	+7.0	+240.0 ²⁾

Table 1: X20 modules - overview for calculating the power output table (cont.)

- 1) For modules with 0.01 W power requirement, the embedded parameter chip can only be read if the I/O supply is also present. The embedded parameter chip is described in chapter 2 "System characteristics", section 9 "Embedded parameter chip" on page 40.
- 2) Rated power at 24 VDC and 10.0 A.

Note: Please observe the example calculations on the following pages.

18. Example 1

Calculation of the power output table for the bus and 24 VDC I/O supply with the following hardware configuration:

Module	Bus power [W]	Internal I/O power [W]	External I/O power [W]	Sensor/actuator supply [W] ¹⁾
DI4371	0.15	0.52	-	12
DI2371	0.12	0.26	-	12
DO4322	0.15	0.31	48 ²⁾	12
DO4322	0.15	0.31	48 ²⁾	12
BT9100	0.45	-	-	-
Subtotal		1.4	96	48
Sum	1.02	145.40		

1) Rated power at 24 VDC and 0.5 A.

2) Rated power at 24 VDC and 100 % simultaneous.

The total power to be supplied by the 24 VDC I/O supply is 145.4 W. One supply module is already integrated in the BR9300 bus receiver. The power comparison indicates that the power provided by the supply module is sufficient.

	Bus power [W]	24 VDC I/O supply power [W]
BR9300	+7.0	+240.0 ¹⁾
Power required by I/O modules	-1.02	-145.4
Power required by all bus modules	-0.78	-
Residual power	+5.2	+94.6

1) Rated power at 24 VDC and 10.0 A.

19. Example 2

In this example, the I/O modules are separated into three potential groups:

- Potential group 1: digital input modules
- Potential group 2: digital output modules
- Potential group 3: analog input modules and temperature modules

Calculation of the power output table for the bus and 24 VDC I/O supply for each potential group with the following hardware configuration:

Potential group 1				
Module	Bus power [W]	Internal I/O power [W]	External I/O power [W]	Sensor/actuator supply [W] ¹⁾
DI6371	0.16	0.78	-	-
DI6371	0.16	0.78	-	-
DI2377	0.2	0.74	-	12
Subtotal		2.30	-	12
Sum	0.52	14.3		

1) Rated power at 24 VDC and 0.5 A.

Potential group 2				
Module	Bus power [W]	Internal I/O power [W]	External I/O power [W] ¹⁾	Sensor/actuator supply [W] ²⁾
DO2322	0.14	0.2	24	12
DO6322	0.16	0.5	72	-
DO8332	0.21	-	- 3)	-
Subtotal		0.7	96	12
Sum	0.51	108.7		

1) Rated power at 24 VDC and 100 % simultaneous.

2) Rated power at 24 VDC and 0.5 A.

3) The power supply is integrated in the module.

Potential group 3				
Module	Bus power [W]	Internal I/O performance [W]	External I/O performance [W]	Sensor/actuator supply [W]
AI4622	0.01	1.1	-	-
AI4622	0.01	1.1	-	-
AT4222	0.01	1.1	-	-
AT2402	0.01	0.5	-	-
BT9100	0.45	-	-	-
Subtotal		3.8	-	-
Sum	0.49	3.8		

Mechanical and electrical configuration • Example 2

Next, a power comparison must be made between the power required by the I/O modules and the power delivered by the supply modules.

Potential group 1 is supplied by the supply module integrated in the BR9300 bus receiver. The total power to be supplied by all the bus modules is 3.34 W. The total power to be supplied by the 24 VDC I/O supply for potential group 1 is 14.3 W.

The power comparison indicates that the power provided by the supply module integrated in the BR9300 is sufficient.

Potential group 1	Bus power [W]	24 VDC I/O supply power [W]
BR9300	+7.0	+240.0 ¹⁾
Power required by I/O modules	-1.52 ²⁾	-14.3 ³⁾
Power required by all bus modules	-1.82	-
Residual power	+3.66	+225.7

1) Rated power at 24 VDC and 10.0 A.

2) Bus power to be supplied for all I/O modules.

3) 24 VDC I/O supply to be provided for potential group 1.

In potential groups 2 and 3, the 24 VDC I/O supply is fed via the PS2100 supply module. A supply module is required for each potential group.

The power comparison indicates that the power provided by the PS2100 is sufficient.

Potential group 2		24 VDC I/O supply power [W]
PS2100		+240.0 ¹⁾
Power required by I/O modules		-108.7
Residual power		+131.3

1) Rated power at 24 VDC and 10.0 A.

Potential group 3		24 VDC I/O supply power [W]
PS2100		+240.0 ¹⁾
Power required by I/O modules		-3.8
Residual power		+236.2

1) Rated power at 24 VDC and 10.0 A.

Chapter 4 • X20 System modules

1. X20 System - module overview

For better orientation and quicker reference, the following lists present the modules in two different arrangements:

- Overview 1: The modules are arranged in ascending alphabetical order
- Overview 2: The modules are arranged according to their module group

1.1 Overview 1: Modules in alphabetical order

Model number	Short description	Page
X20AI2622	X20 analog input module, 2 inputs, $\pm 10\text{ V} / 0$ to 20 mA, 12-bit resolution, configurable input filter	265
X20AI2632	X20 analog input module, 2 inputs, $\pm 10\text{ V} / 0$ to 20 mA, 16-bit resolution, configurable input filter	273
X20AI4622	X20 analog input module, 4 inputs, $\pm 10\text{ V} / 0$ to 20 mA, 12-bit resolution, configurable input filter	279
X20AI4632	X20 analog input module, 4 inputs, $\pm 10\text{ V} / 0$ to 20 mA, 16-bit resolution, configurable input filter	287
X20AO2622	X20 analog output module, 2 outputs, $\pm 10\text{ V} / 0$ to 20 mA, 12-bit resolution	294
X20AO2632	X20 analog output module, 2 outputs, $\pm 10\text{ V} / 0$ to 20 mA, 16-bit resolution	300
X20AO4622	X20 analog output module, 4 outputs, $\pm 10\text{ V} / 0$ to 20 mA, 12-bit resolution	306
X20AO4632	X20 analog output module, 4 outputs, $\pm 10\text{ V} / 0$ to 20 mA, 16-bit resolution	312
X20AT2222	X20 temperature input module, 2 resistance measurement inputs, PT100, PT1000, resolution 0.1 K, 3-wire connections	319
X20AT2402	X20 temperature input module, 2 thermocouple inputs, type J,K,S, resolution 0.1 K	339
X20AT4222	X20 temperature input module, 4 resistance measurement inputs, PT100, PT1000, resolution 0.1 K, 3-wire connections	329
X20AT6402	X20 temperature input module, 6 thermocouple inputs, type J,K,S, resolution 0.1 K	349
X20BB80	Basic X20 bus controller, for bus controller fieldbus interface and bus controller supply module, X20 connection	119
X20BC0043	X20 bus controller fieldbus interface, 1 CANopen interface, status indicator LEDs, order 1 x TB2105 terminal block separately	79
X20BC0053	X20 bus controller fieldbus interface, 1 DeviceNet interface, status indicator LEDs, order 1 x TB2105 terminal block separately	88
X20BC0063	X20 bus controller fieldbus interface, 1 Profibus DP interface, status indicator LEDs, 9-pin DSUB connection	98
X20BC0073	X20 bus controller fieldbus interface, 1 CAN I/O interface, status indicator LEDs, order 1 x TB2105 terminal block separately	105
X20BC0083	X20 bus controller fieldbus interface, ETHERNET Powerlink interface, integrated 2x hub, status indicator LEDs, 2 x RJ45 connection	112

Table 2: Overview 1 - The modules are arranged in ascending alphabetical order

X20 System modules • X20 System - module overview

Model number	Short description	Page
X20BM01	X20 supply bus module, internal I/O supply is split to the left	71
X20BM11	X20 standard bus module, internal I/O supply is interconnected	73
X20BR9300	X20 bus receiver (X2X Link) with feed for internal I/O supply, and X2X Link bus supply	129
X20BT9100	X20 bus transmitter (X2X Link)	136
X20DC1196	X20 digital counter module, 1 channel ABR, 5 V, 250 kHz input freq., 4x evaluation	360
X20DC1198	X20 digital counter module, 1 channel SSI, 5 V, 1 MBit/s, 32-bit	366
X20DC1396	X20 digital counter module, 1 channel ABR, 24 V, 100 kHz input freq., 4x evaluation	372
X20DC1398	X20 digital counter module, 1 channel SSI, 24 V, 125 kBit/s, 32-bit	378
X20DC2395	X20 digital counter module, 1 x SSI absolute encoder (24 V), 1 x ABR incremental encoder (24 V), 2 x AB incremental encoder (24 V), 4 x event counter, or 2 x PWM	395
X20DC2396	X20 digital counter module, 2 ABR channels, 24 V, 100 kHz input freq., 4x evaluation	383
X20DC2398	X20 digital counter module, 2 SSI channels, 24 V, 125 kBit/s, 32-bit	390
X20DC4395	X20 digital counter module, 2 x SSI absolute encoder (24 V), 2 x ABR incremental encoder (24 V), 4 x AB incremental encoder (24 V), 8 x event counter, or 4 x PWM	405
X20DI2371	X20 digital input module, 2 inputs, 24 VDC, sink, configurable input filter, 3-wire connections	158
X20DI2377	X20 digital input module, 2 inputs, 24 VDC, sink, configurable input filter, 2 event counters 50 kHz	164
X20DI4371	X20 digital input module, 4 inputs, 24 VDC, sink, configurable input filter, 3-wire connections	172
X20DI6371	X20 digital input module, 6 inputs, 24 VDC, sink, configurable input filter, 2-wire connections	178
X20DI6372	X20 digital input module, 6 inputs, 24 VDC, source, configurable input filter, 2-wire connections	184
X20DI9371	X20 digital input module, 12 inputs, 24 VDC, sink, configurable input filter, 1-wire connections	190
X20DI9372	X20 digital input module, 12 inputs, 24 VDC, source, configurable input filter, 1-wire connections	196
X20DO2322	X20 digital output module, 2 outputs, 24 VDC, 0.5 A, source, 3-wire connections	203
X20DO4322	X20 digital output module, 4 outputs, 24 VDC, 0.5 A, source, 3-wire connections	210
X20DO4332	X20 digital output module, 4 outputs, 24 VDC, 2.0 A, source, 3-wire connections	217
X20DO6321	X20 digital output module, 6 outputs, 24 VDC, 0.5 A, sink, 2-wire connections	225
X20DO6322	X20 digital output module, 6 outputs, 24 VDC, 0.5 A, source, 2-wire connections	231
X20DO8332	X20 digital output module, 8 outputs, 24 VDC, 2.0 A, source, feed directly on module, 1-wire connections	237
X20DO9321	X20 digital output module, 12 outputs, 24 VDC, 0.5 A, sink, 1-wire connections	246
X20DO9322	X20 digital output module, 12 outputs, 24 VDC, 0.5 A, source, 1-wire connections	252
X20PS2100	X20 supply module for internal I/O supply	144
X20PS3300	X20 supply module for internal I/O supply, X2X link supply	149
X20PS9400	X20 supply module for bus controller and internal I/O supply, X2X link bus supply	121
X20TB06	Standard X20 terminal block (6-pin)	75
X20TB12	Standard X20 terminal block (12-pin)	75
X20ZF0000	Dummy X20 module (non-functional)	416

Table 2: Overview 1 - The modules are arranged in ascending alphabetical order (cont.)

1.2 Overview 2: Modules arranged by module group

Model number	Short description	Page
Bus modules		
X20BM01	X20 supply bus module, internal I/O supply is split to the left	71
X20BM11	X20 standard bus module, internal I/O supply is interconnected	73
Terminal blocks		
X20TB06	Standard X20 terminal block (6-pin)	75
X20TB12	Standard X20 terminal block (12-pin)	75
Bus controller		
X20BC0043	X20 bus controller fieldbus interface, 1 CANopen interface, status indicator LEDs, order 1 x TB2105 terminal block separately	79
X20BC0053	X20 bus controller fieldbus interface, 1 DeviceNet interface, status indicator LEDs, order 1 x TB2105 terminal block separately	88
X20BC0063	X20 bus controller fieldbus interface, 1 Profibus DP interface, status indicator LEDs, 9-pin DSUB connection	98
X20BC0073	X20 bus controller fieldbus interface, 1 CAN I/O interface, status indicator LEDs, order 1 x TB2105 terminal block separately	105
X20BC0083	X20 bus controller fieldbus interface, ETHERNET Powerlink interface, integrated 2x hub, status indicator LEDs, 2 x RJ45 connection	112
Bus controller system modules		
X20BB80	Basic X20 bus controller, for bus controller fieldbus interface and bus controller supply module, X20 connection	119
X20PS9400	X20 supply module for bus controller and internal I/O supply, X2X link bus supply	121
Bus receivers and bus transmitters		
X20BR9300	X20 bus receiver (X2X Link) with feed for internal I/O supply, and X2X Link bus supply	129
X20BT9100	X20 bus transmitter (X2X Link)	136
Supply modules		
X20PS2100	X20 supply module for internal I/O supply	144
X20PS3300	X20 supply module for internal I/O supply, X2X link supply	149
Digital input modules		
X20DI2371	X20 digital input module, 2 inputs, 24 VDC, sink, configurable input filter, 3-wire connections	158
X20DI2377	X20 digital input module, 2 inputs, 24 VDC, sink, configurable input filter, 2 event counters 50 kHz	164
X20DI4371	X20 digital input module, 4 inputs, 24 VDC, sink, configurable input filter, 3-wire connections	172
X20DI6371	X20 digital input module, 6 inputs, 24 VDC, sink, configurable input filter, 2-wire connections	178
X20DI6372	X20 digital input module, 6 inputs, 24 VDC, source, configurable input filter, 2-wire connections	184
X20DI9371	X20 digital input module, 12 inputs, 24 VDC, sink, configurable input filter, 1-wire connections	190
X20DI9372	X20 digital input module, 12 inputs, 24 VDC, source, configurable input filter, 1-wire connections	196

Table 3: Overview 2 - The modules are arranged according to their module group

X20 System modules • X20 System - module overview

Model number	Short description	Page
Digital output modules		
X20DO2322	X20 digital output module, 2 outputs, 24 VDC, 0.5 A, source, 3-wire connections	203
X20DO4322	X20 digital output module, 4 outputs, 24 VDC, 0.5 A, source, 3-wire connections	210
X20DO4332	X20 digital output module, 4 outputs, 24 VDC, 2.0 A, source, 3-wire connections	217
X20DO6321	X20 digital output module, 6 outputs, 24 VDC, 0.5 A, sink, 2-wire connections	225
X20DO6322	X20 digital output module, 6 outputs, 24 VDC, 0.5 A, source, 2-wire connections	231
X20DO8332	X20 digital output module, 8 outputs, 24 VDC, 2.0 A, source, feed directly on module, 1-wire connections	237
X20DO9321	X20 digital output module, 12 outputs, 24 VDC, 0.5 A, sink, 1-wire connections	246
X20DO9322	X20 digital output module, 12 outputs, 24 VDC, 0.5 A, source, 1-wire connections	252
Analog input modules		
X20AI2622	X20 analog input module, 2 inputs, $\pm 10\text{ V} / 0$ to 20 mA, 12-bit resolution, configurable input filter	265
X20AI2632	X20 analog input module, 2 inputs, $\pm 10\text{ V} / 0$ to 20 mA, 16-bit resolution, configurable input filter	273
X20AI4622	X20 analog input module, 4 inputs, $\pm 10\text{ V} / 0$ to 20 mA, 12-bit resolution, configurable input filter	279
X20AI4632	X20 analog input module, 4 inputs, $\pm 10\text{ V} / 0$ to 20 mA, 16-bit resolution, configurable input filter	287
Analog output modules		
X20AO2622	X20 analog output module, 2 outputs, $\pm 10\text{ V} / 0$ to 20 mA, 12-bit resolution	294
X20AO2632	X20 analog output module, 2 outputs, $\pm 10\text{ V} / 0$ to 20 mA, 16-bit resolution	300
X20AO4622	X20 analog output module, 4 outputs, $\pm 10\text{ V} / 0$ to 20 mA, 12-bit resolution	306
X20AO4632	X20 analog output module, 4 outputs, $\pm 10\text{ V} / 0$ to 20 mA, 16-bit resolution	312
Temperature modules		
X20AT2222	X20 temperature input module, 2 resistance measurement inputs, PT100, PT1000, resolution 0.1 K, 3-wire connections	319
X20AT4222	X20 temperature input module, 4 resistance measurement inputs, PT100, PT1000, resolution 0.1 K, 3-wire connections	329
X20AT2402	X20 temperature input module, 2 thermocouple inputs, type J,K,S, resolution 0.1 K	339
X20AT6402	X20 temperature input module, 6 thermocouple inputs, type J,K,S, resolution 0.1 K	349
Counter modules		
X20DC1196	X20 digital counter module, 1 channel ABR, 5 V, 250 kHz input freq., 4x evaluation	360
X20DC1198	X20 digital counter module, 1 channel SSI, 5 V, 1 MBit/s, 32-bit	366
X20DC1396	X20 digital counter module, 1 channel ABR, 24 V, 100 kHz input freq., 4x evaluation	372
X20DC1398	X20 digital counter module, 1 channel SSI, 24 V, 125 kBit/s, 32-bit	378
X20DC2396	X20 digital counter module, 2 ABR channels, 24 V, 100 kHz input freq., 4x evaluation	383
X20DC2398	X20 digital counter module, 2 SSI channels, 24 V, 125 kBit/s, 32-bit	390
X20DC2395	X20 digital counter module, 1 x SSI absolute encoders 24 V, 1 x ABR incremental encoder 24 V, 2 x AB incremental encoder 24 V, 4 x event counter, or 2 x PWM	395
X20DC4395	X20 digital counter module, 2 x SSI absolute encoder (24 V), 2 x ABR incremental encoder (24 V), 4 x AB incremental encoder (24 V), 8 x event counter, or 4 x PWM	405

Table 3: Overview 2 - The modules are arranged according to their module group (cont.)

Model number	Short description	Page
	Dummy module	
X20ZF0000	Dummy X20 module (non-functional)	416
	Accessories	
	Terminal locking clip, plain text cover, locking plate, printer, etc.	419

Table 3: Overview 2 - The modules are arranged according to their module group (cont.)

2. Bus modules

2.1 General information

In the X20 System, a bus module takes the place of a rack (see chapter 3 "Mechanical and electrical configuration", section 6.1 "Bus module replaces rack" on page 52).

2.1.1 Overview

Model number	Short description
X20BM01	X20 supply bus module, internal I/O supply is split to the left
X20BM11	X20 standard bus module, internal I/O supply is interconnected

Table 4: Overview of bus modules

2.2 BM01

2.2.1 General information

The BM01 bus module is the basis for all X20 supply modules.

- Basis for all X20 supply modules
- For creating electrical potential groups
- The internal I/O supply is interrupted towards the left

2.2.2 Order data

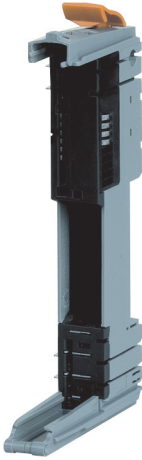
Model number	Short description	Image
	Bus module	
X20BM01	X20 supply bus module, internal I/O supply is split to the left	

Table 5: BM01 - order data

2.2.3 Technical data

Product ID	BM01
Short description	
Bus module	Supply bus module, internal I/O supply is interrupted to the left
General information	
Power consumption Bus I/O internal	Typ. 0.13 W -
Certification	CE, C-UL-US, GOST-R (in development)

Table 6: BM01 - technical data

Product ID	BM01
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 ^{+0.2} mm

Table 6: BM01 - technical data (cont.)

2.2.4 Potential control

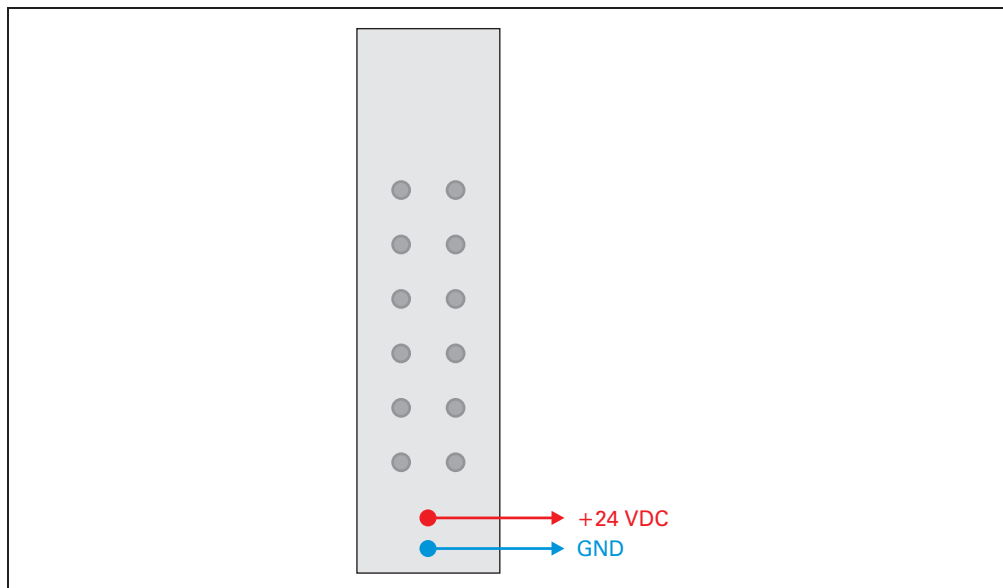


Figure 22: BM01 - potential control

2.3 BM11

2.3.1 General information

The BM1 bus module is the basis for all X20 I/O modules.

- Standard bus module
- The internal I/O supply is interconnected

2.3.2 Order data

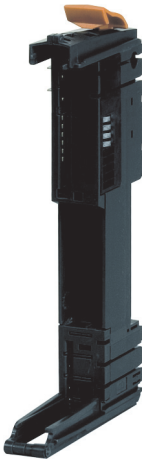
Model number	Short description	Image
	Bus module	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 7: BM11 - order data

2.3.3 Technical data

Product ID	BM11
Short description	
Bus module	Standard bus module, the internal I/O supply is interconnected
General information	
Power consumption Bus I/O internal	Typ. 0.13 W -
Certification	CE, C-UL-US, GOST-R (in development)

Table 8: BM11 - technical data

Product ID	BM11
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm

Table 8: BM11 - technical data (cont.)

2.3.4 Potential control

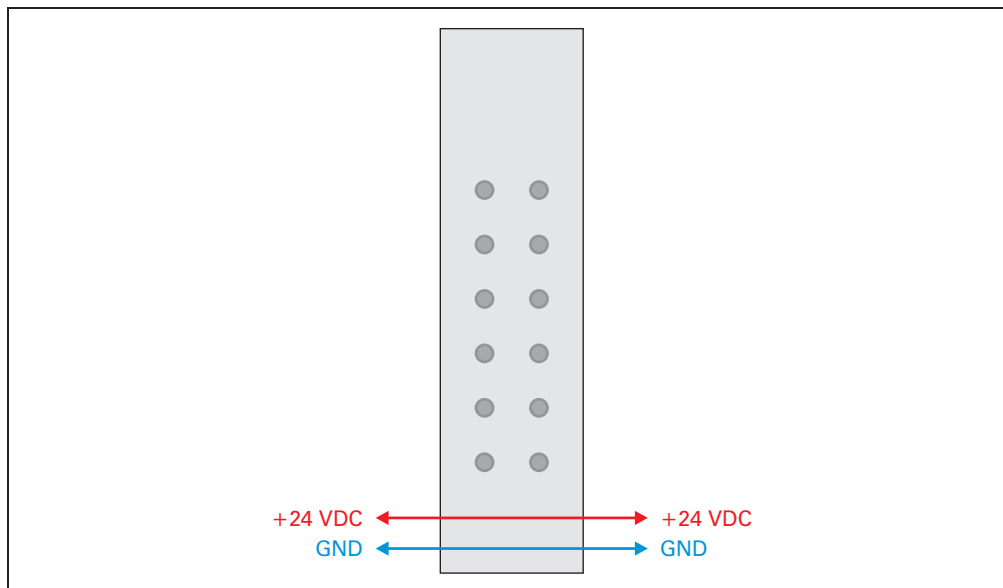


Figure 23: BM11 - potential control

3. Terminal blocks

3.1 6/12-pin terminal blocks TB06 / TB12

3.1.1 General information

The X20 modules are wired using the TB06 and TB12 terminal blocks.

- Tool-free wiring with push-in technology
- Simple wire release using lever
- Ability to label each terminal
- Plain text labeling also possible
- Test access for standard probes
- Potential for customer coding

3.1.2 Order data

Model number	Short description	Image
	Terminal block	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
		X20TB06 X20TB12

Table 9: TB06/TB12 - order data

3.1.3 Technical data

Product ID	TB06/TB12
Short description	
Terminal block 6-pin 12-pin	X20TB06 X20TB12
Terminal block	
Type of terminal	Push-in terminal
Distance between contacts Left - right Above - below	4.2 mm 10.96 mm
Contact resistance	$\leq 5 \text{ m}\Omega$
Rated voltage	230 VAC
Rated current ¹⁾	10 A / contact
Connection cross section solid wire line fine wire line with wire tip sleeves	0.08 mm ² - 2.5 mm ² / AWG 28 - 14 0.25 mm ² - 2.5 mm ² / AWG 24 - 14 0.25 mm ² - 1.5 mm ² / AWG 24 - 16 up to 2 x 0.75 mm ² with double wire tip sleeves
Cable type	Only copper wires (no aluminum wires!)
Note	Rated values according to UL

Table 10: TB06/TB12 - technical data

1) Important - keep in mind the data limits for the I/O modules.

4. Bus controller

4.1 General information

The bus controllers are a continuation of the completely modular strategy used for the I/O slices. Made up of a base module, a supply module to supply the voltage for the entire system and a fieldbus interface, the bus controller is an extremely flexible fieldbus connection.

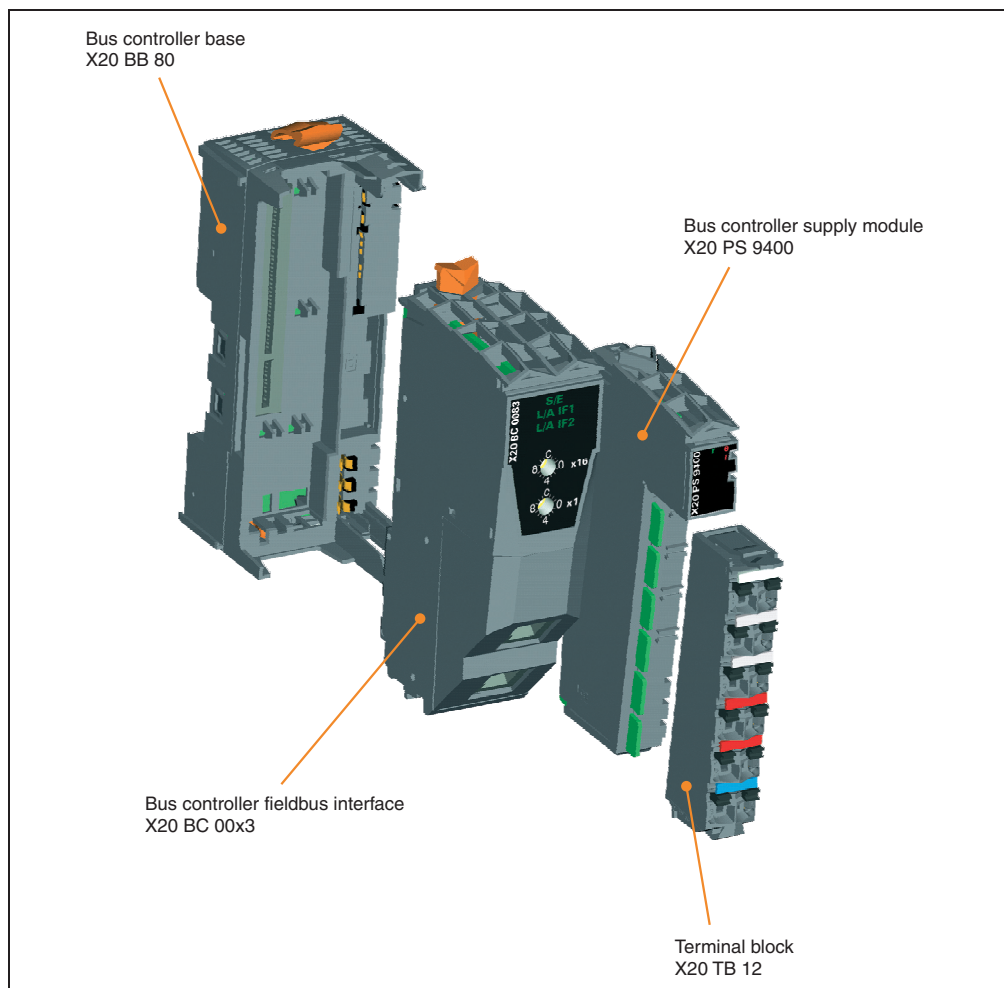


Figure 24: The four parts of a bus controller - fieldbus interface, base module, supply module, terminal block

The entire backplane can be preinstalled. With the removable terminals, the entire system can be wired separately from the electronics module. The individual modules are put in place during commissioning. This is where the I/O system is adapted to the fieldbus being used.

Unlike the Compact CPU with integrated fieldbus connection, the bus controller does not need to be programmed in order to transfer or receive the I/O data on the fieldbus. It can be configured on the fieldbus master.

4.1.1 Overview

Model number	Short description
X20BC0043	X20 bus controller fieldbus interface, 1 CANopen interface, status indicator LEDs, order 1 x TB2105 terminal block separately
X20BC0053	X20 bus controller fieldbus interface, 1 DeviceNet interface, status indicator LEDs, order 1 x TB2105 terminal block separately
X20BC0063	X20 bus controller fieldbus interface, 1 Profibus DP interface, status indicator LEDs, 9-pin DSUB connection
X20BC0073	X20 bus controller fieldbus interface, 1 CAN I/O interface, status indicator LEDs, order 1 x TB2105 terminal block separately
X20BC0083	X20 bus controller fieldbus interface, ETHERNET Powerlink interface, integrated 2x hub, status indicator LEDs, 2 x RJ45 connection

Table 11: Bus controller overview

4.2 BC0043

4.2.1 General information

Use of CAN (Controller Area Network) has considerably expanded in automation technology. CAN topology is based on a line structure and uses twisted pair wires for data transfer. CANopen is a higher layer protocol based on CAN. This standardized protocol offers highly flexible configuration possibilities.

The BC0043 bus controller makes it possible to connect X2X Link I/O nodes to CANopen. It has automatic transfer rate detection and AutoMapping of the I/O modules connected with X2X Link. All CANopen operating modes such as synchronous, event, and polling modes are supported together with PDO linking, life/node guarding, emergency objects, and much more.

- Fieldbus: CANopen
- I/O configuration via the fieldbus
- 20 Receive PDOs and 20 Transmit PDOs
- Select between entry of a fixed transfer rate or automatic transfer rate detection.
- Integrated terminating resistance

4.2.2 Order data

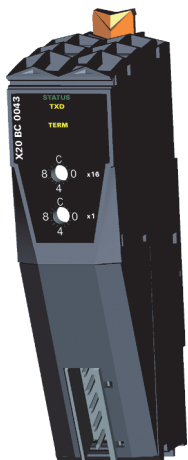
Model number	Short description	Image
	Bus controller	
X20BC0043	X20 bus controller fieldbus interface, 1 CANopen interface, status indicator LEDs, order 1 x TB2105 terminal block separately	
	Required accessories	
0TB2105.9010	Accessory, terminal block, 5-pin, screw clamps, 2.5 mm ²	
0TB2105.9110	Accessory, terminal block, 5-pin, cage clamps, 2.5 mm ²	
X20TB12	Standard X20 terminal block (12-pin)	
X20PS9400	X20 supply module for bus controller and internal I/O supply, X2X link bus supply	
X20BB80	Basic X20 bus controller, for bus controller fieldbus interface and bus controller supply module, X20 connection	

Table 12: BC0043 - order data

4.2.3 Technical data

Product ID	BC0043
Short description	
Bus controller	CANopen
Fieldbus	
Type	CANopen
Type	5-pin multipoint connector
Maximum distance	1,000 m
Maximum transfer rate	1 MBit/s
Determination of transfer rate	Automatic transfer rate detection or fixed rate setting
General information	
Status indicators	Module status, bus function, data transfer, terminating resistance
Diagnostics Module status Bus function Data transfer Terminating resistors	Yes, with status LED and software status Yes, with status LED Yes, with status LED Yes, with status LED
Electrical isolation Fieldbus - X2X bus Fieldbus - I/O	No Yes
Power consumption - bus	1.5 W
Certification	CE, C-UL-US (in development), GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	37.5 ±0.2 mm
Note	Order standard terminal block 1 x TB2105 separately Order standard terminal block 1 x X20TB12 separately Order supply module 1 x X20PS9400 separately Order bus controller base 1 x X20BB80 separately

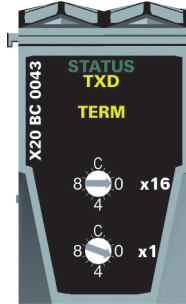
Table 13: BC0043 - technical data

4.2.4 Additional technical data

Product ID	BC0043
Fieldbus	
Controller	SJA 1000 controller
Processor	32-bit RISC processor
Memory	512 KB
Terminating resistance	Integrated in the module
General information	
B&R ID code	\$1F1A

Table 14: BC0043 - additional technical data

4.2.5 Status LEDs



LED	Color	Status	Description
STATUS ¹⁾	Green	Blinking (200 ms on / 200 ms off)	Preoperational mode
		On	RUN mode
		Blinking (200 ms on / 1 s off)	Stop mode
		Blinking (3 ms on / 500 ms off)	Flash delete in progress
	Red	Blinking (200 ms on / 1 s off)	Bus errors
		On	Bus errors: Bus is off
		Blinks 3 x and then is off for 1 s	Transfer rate selection
		Blinking (400 ms on / 800 ms off)	Invalid node switch position
	Blinking green / red		Transfer rate detection in progress
TXD	Yellow	Off	The bus controller is not transmitting any data via the CANopen fieldbus
		On	The bus controller is transmitting data via CANopen fieldbus
TERM	Yellow	Off	The terminating resistance integrated in the bus controller is turned off
		On	The terminating resistance integrated in the bus controller is turned on

Table 15: BC0043 - status indicators

1) The STATUS LED is a green/red dual LED.

4.2.6 Operating and connection elements

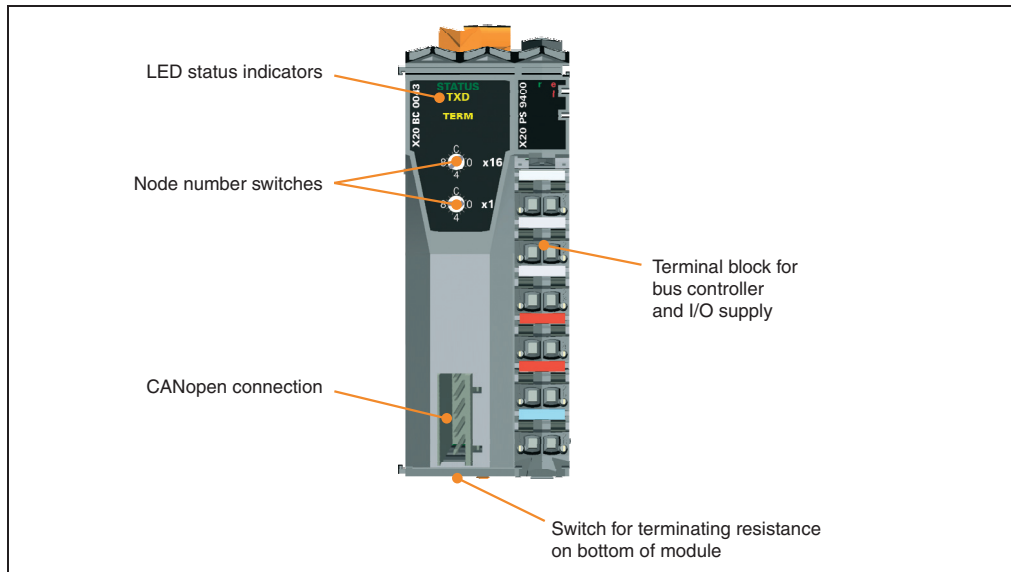


Figure 25: BC0043 - operational and connection elements

4.2.7 CAN interface

The CAN interface is a 5-pin multipoint connector. The TB2105 terminal block must be ordered separately.

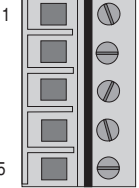
5-pin terminal block	Terminal	Assignment	
 TB2105	1	CAN_L	CAN ground
	2	CAN_L	CAN low
	3	SHLD	Shield
	4	CAN_H	CAN high
	5	Reserved	

Table 16: BC0043 - pin assignments - CAN interface

4.2.8 Terminating resistors

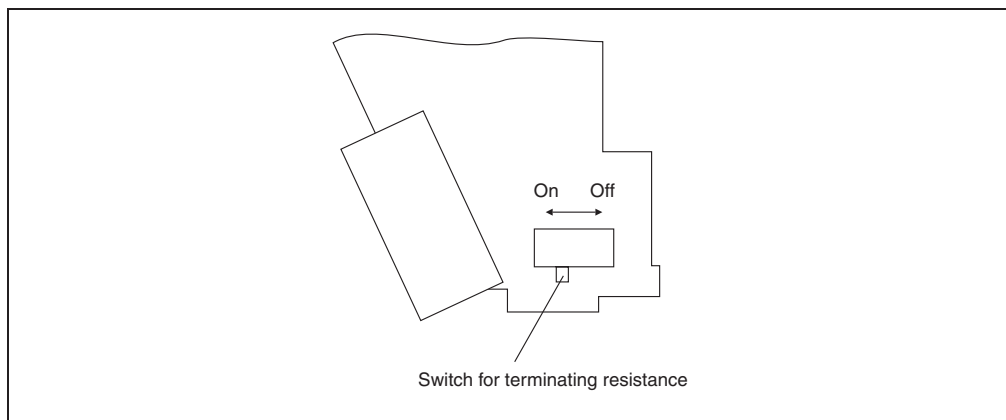


Figure 26: BC0043 - terminating resistance

A terminating resistance is already integrated on the bus controller BC0043. It is turned on and off with a switch on the bottom of the housing. An active terminating resistance is indicated by the TERM LED.

4.2.9 Node numbers and transfer rate

Node numbers and transfer rates are configured using both bus controller number switches.

The transfer rate can be defined in two ways:

- Automatic detection by the bus controller (see section 4.2.10 "Automatic transfer rate detection" on page 85)
- Fixed setting by the user (see section 4.2.11 "Setting a fixed transfer rate" on page 86)

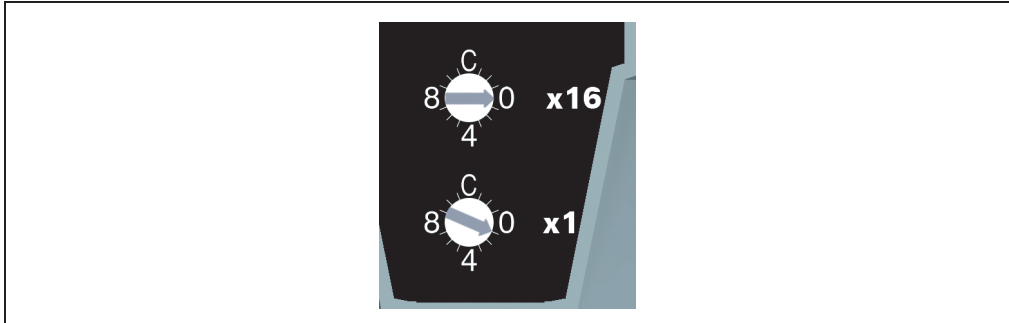


Figure 27: BC0043 - number switches for node numbers and transfer rates

Switch position	Node number	Transfer rate
\$00	Not allowed	-
\$01 - \$7F	1 - 127	Automatically set by the bus controller (default) or fixed setting by the user.
\$80 - \$89	-	Setting a fixed transfer rate
\$8A - \$8F	Not allowed	-
\$90	Delete parameter See section 4.2.12 "Delete parameter" on page 87	-
\$91 - \$FF	Not allowed	-

Table 17: BC0043 - node numbers and transfer rates

4.2.10 Automatic transfer rate detection

After booting, the bus controller BC0043 goes into "Listen Only" mode. That means the bus controller behaves passively on the bus and only listens.

The BC0043 tries to receive valid objects. If errors occur during receiving, the controller switches to the next transfer rate from the search table.

If no objects are received, all transfer rates are tested cyclically. This procedure is repeated until valid objects are received.

Search table

The controller tests the transfer rate according to this table. Beginning with the starting transfer rate (1000 kBit/s), the controller switches to the next lower transfer rate. At the end of the table, the bus controller starts searching from the beginning again.

Transfer rate
1000 kBit/sec
500 kBit/s
250 kBit/s
125 kBit/s
50 kBit/s
20 kBit/s
10 kBit/s

Table 18: BC0043 - transfer rate search table

4.2.11 Setting a fixed transfer rate

The default setting for the bus controller is automatic transfer rate detection. It is also possible to set a fixed transfer rate using switch positions \$80 - \$89.

Switch position	Transfer rate
\$80	1000 kBit/s
\$81	800 kBit/s
\$82	500 kBit/s
\$83	250 kBit/s
\$84	125 kBit/s
\$85	100 kBit/s
\$86	50 kBit/s
\$87	20 kBit/s
\$88	10 kBit/s
\$89	Automatic transfer rate detection

Table 19: BC0043 - possible fixed transfer rates

Programming a fixed transfer rate

- 1) Turn off power supply to the bus controller
- 2) Define transfer rate to be programmed by setting the node numbers (\$80 - \$89)
- 3) Turn power supply to the bus controller back on
- 4) Wait until the STATUS LED blinks with a red triple-flash (transfer rate is now programmed).
- 5) Turn off power supply to the bus controller
- 6) Set desired node number (\$01 - \$7F)
- 7) Turn power supply to the bus controller back on
- 8) Bus controller boots with the set node number and the programmed transfer rate

4.2.12 Delete parameter

Multiple parameters can be stored in the bus controller flash memory:

- Communication parameters
- Application parameters
- Fixed transfer rates

Deleting the parameters using the switch position \$90 returns the bus controller to its factory settings.

Deleting the parameters listed above

- 1) Turn off power supply to the bus controller
- 2) Set the node number to \$90
- 3) Turn power supply to the bus controller back on
- 4) Wait until the STATUS LED blinks green for 5 s (100 ms on / 200 ms off). The node number switch must be set to \$00 and then back to \$90 within this time window.
- 5) Wait until the STATUS LED blinks with a red double-flash (parameters are now deleted).
- 6) Turn off power supply to the bus controller
- 7) Set desired node number (\$01 - \$7F)
- 8) Turn power supply to the bus controller back on
- 9) Bus controller boots with the set node number and automatic transfer rate detection

4.2.13 Further documentation and import files (EDS)

Further documentation about CANopen bus controller functions, as well as the necessary import files for the master engineering tool, are available for download from the B&R Homepage (www.br-automation.com).

4.3 BC0053

4.3.1 General information

DeviceNet was developed by Allen Bradley as a CAN bus based automation network. It is based on a producer/consumer protocol. From the user's point of view, data is handled completely separately from CAN bus transfer possibilities (e.g. longer data packets are automatically fragmented by DeviceNet). Access occurs using I/O messages with defined properties.

The BC0053 bus controller makes it possible to connect X2X Link I/O nodes to DeviceNet. It has automatic transfer rate detection and AutoMapping of the I/O modules connected with X2X Link. Explicit messaging, change of state, cyclic, polled and bit strobe are supported as DeviceNet operating modes. In addition to the standard communication objects, there are also a number of manufacturer-specific objects.

X20 or other modules that are based on X2X Link can be connected to the bus controller. The entire configuration of this type of modular system is supported by the DeviceNet standard. Allen Bradley developed this modular I/O configuration to simplify the necessary configuration steps and to achieve the required configuration of a modular DeviceNet device intuitively on a flat and very user-friendly interface. The X20 DeviceNet bus controller from B&R also supports this type of configuration.

- Fieldbus: DeviceNet
- I/O configuration via the fieldbus
- Support of both linear and modular systems (Allen Bradley)
- Integrated terminating resistance

4.3.2 Order data

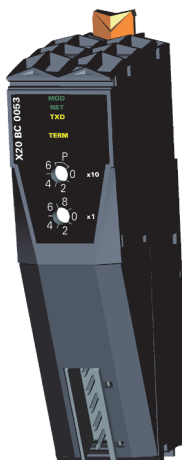
Model number	Short description	Image
	Bus controller	
X20BC0053	X20 bus controller fieldbus interface, 1 DeviceNet interface, status indicator LEDs, order 1 x TB2105 terminal block separately	
	Required accessories	
0TB2105.9010	Accessory, terminal block, 5-pin, screw clamps, 2.5 mm ²	
0TB2105.9110	Accessory, terminal block, 5-pin, cage clamps, 2.5 mm ²	
X20TB12	Standard X20 terminal block (12-pin)	
X20PS9400	X20 supply module for bus controller and internal I/O supply, X2X link bus supply	
X20BB80	Basic X20 bus controller, for bus controller fieldbus interface and bus controller supply module, X20 connection	

Table 20: BC0053 - order data

4.3.3 Technical data

Product ID	BC0053
Short description	
Bus controller	DeviceNet
Fieldbus	
Type	DeviceNet
Type	5-pin multipoint connector
Maximum distance	500 m at 125 kBit/sec
Maximum transfer rate	500 kBit/s
Determination of transfer rate	Automatic transfer rate detection
General information	
Status indicators	Module status, bus function, 24 V DeviceNet voltage, data transfer, terminating resistance.
Diagnostics Module status Bus function 24 V DeviceNet voltage Data transfer Terminating resistors	Yes, with status LED and software status Yes, with status LED Yes, with status LED Yes, with status LED Yes, with status LED
Electrical isolation Fieldbus - X2X bus Fieldbus - I/O	No Yes

Table 21: BC0053 - technical data

Product ID	BC0053
Power consumption - bus	1.5 W
Certification	CE, C-UL-US (in development), GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	37.5 +0.2 mm
Note	Order standard terminal block 1 x TB2105 separately Order standard terminal block 1 x X20TB12 separately Order supply module 1 x X20PS9400 separately Order bus controller base 1 x X20BB80 separately

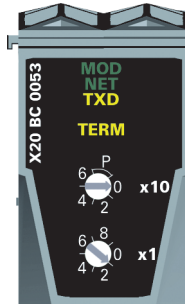
Table 21: BC0053 - technical data (cont.)

4.3.4 Additional technical data

Product ID	BC0053
Fieldbus	
Controller	SJA 1000 controller
Processor	32-bit RISC processor
Memory	512 KB
Terminating resistance	Integrated in the module
General information	
B&R ID code	\$1F1B

Table 22: BC0053 - additional technical data

4.3.5 Status LEDs



LED	Color	Status	Description
MOD ¹⁾	Green	Off	Bus sense error: If the NET LED is also out, there is no 24 V DeviceNet voltage. No transfer rate: If the PS9400's RUN LED is active (Preoperational or RUN mode), the automatic transfer rate detection is still running or no transfer rate could be detected.
		On	RUN mode: The 24 V DeviceNet voltage is OK and the module is operating under normal conditions.
		Blinking	Standby mode: Configuration is missing, incomplete, or incorrect.
	Red	Blinking	Recoverable Fault mode:
	Blinking green / red		Module is performing a self test.

Table 23: BC0053 - status indicators

LED	Color	Status	Description
NET ¹⁾	Green	Off	No power, offline: • Bus sense error: If the MOD LED is also out, there is no 24 V DeviceNet voltage. • No transfer rate: If the PS9400's RUN LED is active (Preoperational or RUN mode), the automatic transfer rate detection is still running or no transfer rate could be detected. • Module has not yet completed a duplicate MAC-ID test.
		Blinking	Online, not connected: • The module has carried out the duplicate MAC-ID test and is online. • There is no established connection to a master/scanner.
		On	Everything is OK: A connection to the master/scanner (explicit or I/O) is set up.
	Red	Blinking	Connection timeout: The timeout for an I/O connection has expired.
		On	Critical connection error - fieldbus communication no longer possible: • Duplicate MAC ID error • Bus off • Receive/transmit overrun
TXD	Yellow	Off	The bus controller is not transmitting any data via the DeviceNet fieldbus
		On	The bus controller is transmitting data via DeviceNet fieldbus
TERM	Yellow	Off	The terminating resistance integrated in the bus controller is turned off
		On	The terminating resistance integrated in the bus controller is turned on

Table 23: BC0053 - status indicators (cont.)

1) The MOD and NET LEDs are green/red dual LEDs.

4.3.6 Operating and connection elements

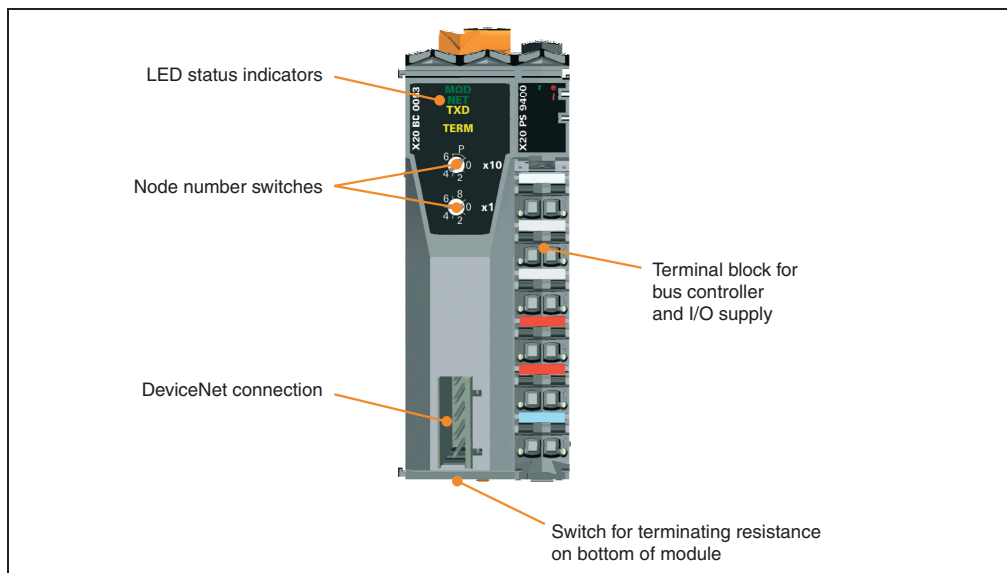


Figure 28: BC0053 - operational and connection elements

4.3.7 DeviceNet Interface

The DeviceNet interface is a 5-pin multipoint connector. The TB2105 terminal block must be ordered separately.

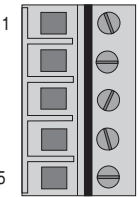
5-pin terminal block	Terminal	Assignment	
 TB2105	1	CAN_L (V-)	CAN ground
	2	CAN_L	CAN low
	3	SHLD	Shield
	4	CAN_H	CAN high
	5	V+	Bus supply

Table 24: BC0053 - pin assignments - DeviceNet interface

4.3.8 Terminating resistors

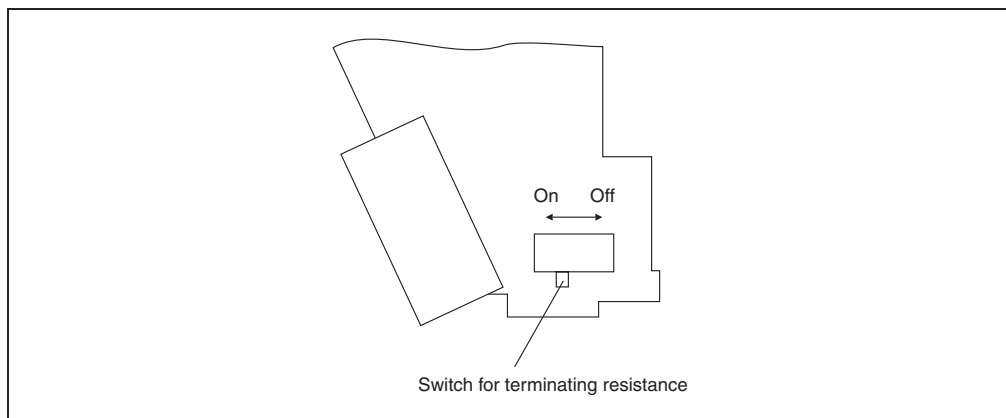


Figure 29: BC0053 - terminating resistance

A terminating resistance is already integrated on the bus controller BC0053. It is turned on and off with a switch on the bottom of the housing. An active terminating resistance is indicated by the TERM LED.

4.3.9 Node number

The MAC-ID is configured using the two address switches on the bus controller.

The adjustable area lies between 0 and 63. This value range is required in the DeviceNet specifications for a DeviceNet unit.

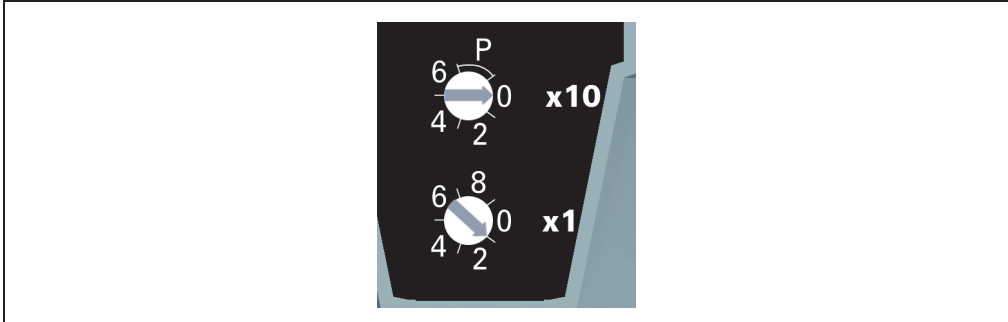


Figure 30: BC0053 - address switches

Switch position	MAC ID
00 - 63	0 - 63
64	Setting MAC-ID using software configuration
65 - 89	Not allowed
90	Delete parameter See section 4.3.11 "Delete parameter" on page 97
91 - 99	Not allowed

Table 25: BC0053 - MAC ID

Special functions

Position of address switch	Special functions
64	The MAC ID can be configured by setting the address switch using the master/scanner software.

Table 26: BC0053 - special functions

4.3.10 Automatic transfer rate detection

After booting, the bus controller BC0053 goes into "Listen Only" mode. That means the bus controller behaves passively on the bus and only listens.

The BC0053 tries to receive valid objects. If errors occur during receiving, the controller switches to the next transfer rate from the search table.

If no objects are received, all transfer rates are tested cyclically. This procedure is repeated until valid objects are received and the correct transfer rate has been defined. Only transfer rates allowed by DeviceNet specifications are tested.

Search table

The controller tests the transfer rate according to this table. Beginning with the starting transfer rate (500 kBit/s), the controller switches to the next lower transfer rate. At the end of the table, the bus controller starts searching from the beginning again.

Transfer rate
500 kBit/s
250 kBit/s
125 kBit/s

Table 27: BC0053 - transfer rate search table

Information:

While automatic transfer rate recognition is running, both DeviceNet LEDs are switched off (because there is no LED status definition for this status according to DeviceNet specifications).

To ensure that the module has been supplied and booted, this manufacturer specific status definition requires that PS9400's RUN LED is active.

4.3.11 Delete parameter

Multiple parameters can be stored in the bus controller flash memory. Deleting the parameters using the switch position 90 returns the bus controller to its factory settings.

Deleting parameters

- 1) Turn off power supply to the bus controller
- 2) Set the node number to 90
- 3) Turn power supply to the bus controller back on
- 4) Wait until the STATUS LED blinks green for 5 s (3 ms on / 500 ms off). The node number switch must be set to 00 and then back to 90 within this time window.
- 5) Wait until the STATUS LED blinks with a red double-flash (parameters are now deleted)
- 6) Turn off power supply to the bus controller
- 7) Set desired node number (00 - 63)
- 8) Turn power supply to the bus controller back on
- 9) Bus controller boots with the set node number and automatic transfer rate detection

4.3.12 Further documentation and import files (EDS)

Further documentation about DeviceNet bus controller functions, as well as the necessary import files for the master engineering tool, are available for download from the B&R Homepage (www.br-automation.com).

4.4 BC0063

4.4.1 General information

Profibus DP is based on the physics of the RS485 interface. Data transfer is controlled using a hybrid bus access procedure. Active stations receive communication rights via a token passing procedure and can then access all stations on the network according to the master-slave principle. The maximum time of circulation for a token can be configured, which results in a defined cycle time.

Access represents various services for the user, for cyclic and for acyclic data transfer.

The BC0063 bus controller makes it possible to connect X2X Link I/O nodes to Profibus DP. It supports Profibus DP with all of its options and other additional properties. In addition to the device, module, and channel diagnostics provided in the Profibus standard, it is also possible e. g. to switch to the slot diagnostics option in S7 format. X20 or other modules that are based on X2X Link can be connected to the bus controller. The modular system configuration is optimally supported by Profibus DP.

- Fieldbus: Profibus DP
- I/O configuration via the fieldbus
- Extensive device, module, and channel diagnosis according to Profibus DP standard
- Communication with X2X Link I/O nodes even works when some nodes are missing or without power

4.4.2 Order data

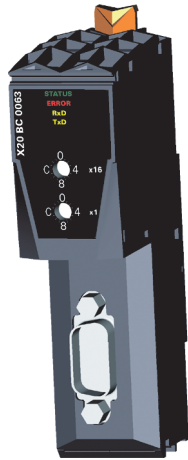
Model number	Short description	Image
	Bus controller	
X20BC0063	X20 bus controller fieldbus interface, 1 Profibus DP interface, status indicator LEDs, 9-pin DSUB connection	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20PS9400	X20 supply module for bus controller and internal I/O supply, X2X link bus supply	
X20BB80	Basic X20 bus controller, for bus controller fieldbus interface and bus controller supply module, X20 connection	

Table 28: BC0063 - order data

4.4.3 Technical data

Product ID	BC0063
Short description	
Bus controller	Profibus DP slave
Fieldbus	
Type	Profibus DP slave
Type	9-pin DSUB socket
Maximum distance	See Profibus DP specifications
Maximum transfer rate	12 MBit/s
Determination of transfer rate	Automatic transfer rate detection
General information	
Status indicators	Module status, bus function, data transfer
Diagnostics	
Module status	Yes, with status LED and software status
Bus function	Yes, with status LED
Data transfer	Yes, with status LED
Electrical isolation	
Fieldbus - X2X bus	No
Fieldbus - I/O	Yes
Power consumption - bus	2.3 W
Certification	CE, C-UL-US (in development), GOST-R (in development)

Table 29: BC0063 - technical data

Product ID	BC0063
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	37.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order supply module 1 x X20PS9400 separately Order bus controller base 1 x X20BB80 separately

Table 29: BC0063 - technical data (cont.)

4.4.4 Additional technical data

Product ID	BC0063
Fieldbus	
Controller	VPC3
Processor	32-bit RISC processor
Memory	512 KB
General information	
B&R ID code	\$1F1C

Table 30: BC0063 - additional technical data

4.4.5 Status LEDs

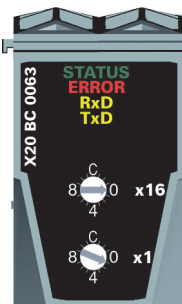
Image	LED	Description		
	STATUS and ERROR	Status indicator for Profibus DP bus controllers.		
		STATUS (green)	ERROR (red)	Description
		Off	Off	HARDWARE FAULT / POWER FAIL
		On	On	BUS OFF
		On	Blinking	WAIT FOR CONFIG
		Blinking	Off	DATA EXCHANGE - DIAGNOSTICS
		On	Off	DATA EXCHANGE - NO ERROR
		Blinking	Blinking	CONFIG ERROR
		Off	Blinking	SERVICE MODE - BOOT
		Single flash	Single flash	HARDWARE FAULT
		For a more detailed description see section 4.4.10 "Diagnostics" on page 104.		
RXD	The yellow RxD LED lights up when the bus controller is receiving data from the Profibus DP fieldbus.			
TxD	The yellow TxD LED lights up when the bus controller is transferring data via the Profibus DP fieldbus.			

Table 31: BC0063 - status indicators

4.4.6 Operating and connection elements

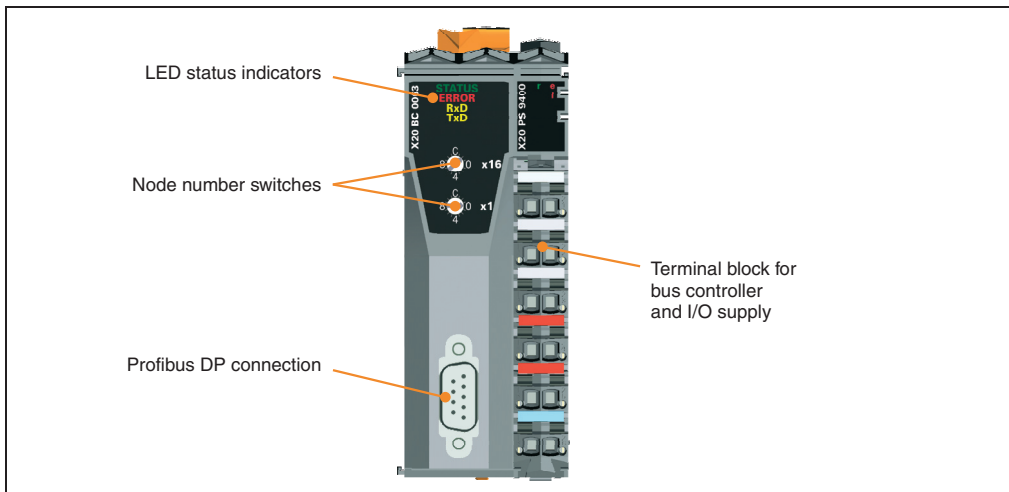


Figure 31: BC0063 - operational and connection elements

4.4.7 Profibus DP interface

The Profibus DP interface is a 9-pin DSUB socket.

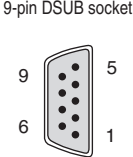
Interface	Pin	RS485		
 9-pin DSUB socket	1	Reserved		
	2	Reserved		
	3	RxD/TxD-P	Data	Red
	4	CNTR-P	Transmit enable	
	5	DGND	Power supply	
	6	VP	Power supply	
	7	Reserved		
	8	RxD/TxD-N	Data\	Green
	9	CNTR-N	Transmit enable\	
CNTR ... Directional switch for external repeater				

Table 32: BC0063 - pin assignments - Profibus DP interface

4.4.8 Node number

The Profibus DP node number is configured using both number switches of the bus controller.

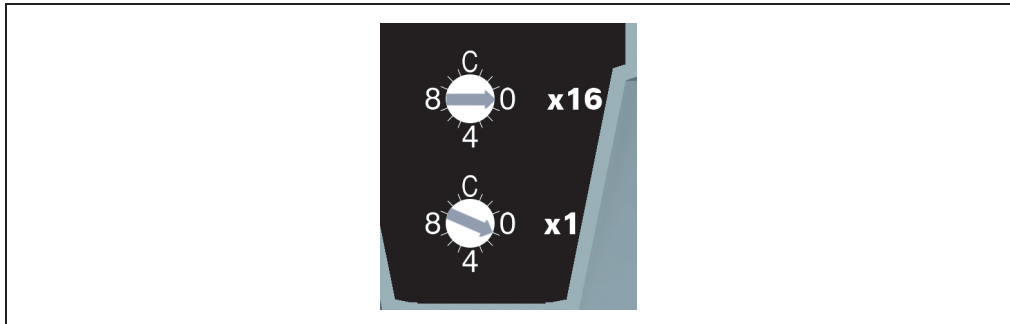


Figure 32: BC0063 - number switch

Switch position	Node number
\$00	Not allowed
\$01 - \$7D	1 - 125
\$7E - \$FF	Not allowed

Table 33: BC0063 - node numbers

4.4.9 Automatic transfer rate detection

After booting or after a communication timeout, the bus controller goes into the status "Baud Search". This means the bus controller behaves passively on the bus.

The bus controller always begins the search for the configured transfer rate with the highest transfer rate. If a complete error-free telegram is not received during monitoring time, then the search is continued using the next lowest transfer rate.

Transfer rate
12 MBit/s
6 MBit/s
3 MBit/s
1.5 MBit/s
500 kBit/s
187.5 kBit/s
93.75 kBit/s
45.45 kBit/s
19.2 kBit/s
9.6 kBit/s

Table 34: BC0063 - transfer rates supported by the bus controller

4.4.10 Diagnostics

The condition of the Profibus DP bus controller is diagnosed using the status LEDs STATUS and ERROR.

STATUS	ERROR	Description	Remedy
Off	Off	HARDWARE FAULT / POWER FAIL	Check wiring of power supply (see section 5.3.7 "Connection examples" on page 125)
On	On	BUS OFF - No transfer rate detected - No connection to the DP master - DP master not active	- Check the Profibus network - Check the Profibus master
On	Blinking	WAIT FOR CONFIG Transfer rate has been detected, but the Profibus master has not yet configured the bus controller	- Check the node number switch - Check the slave address in the master configuration
Blinking	Off	DATA EXCHANGE - DIAGNOSTICS - The bus controller is still initializing the I/O modules - The I/O modules configured by the master cannot be found - An error has occurred on one or more I/O modules (short circuit, etc.)	- Initialization can take a few seconds depending on the number of I/O modules connected - Check the wiring and power supply for the I/O modules - Read diagnostic messages in the respective Profibus master's Engineering Tool
On	Off	DATA EXCHANGE Cyclic data exchange with the Profibus DP master	
Blinking	Blinking	CONFIG ERROR - One or more I/O modules found do not match with the configuration of the Profibus DP master - The configuration received from the Profibus master is invalid	- Check the wiring of the X2X link and the order of I/O modules - Check the configuration of the Profibus master - Read diagnostic messages in the respective Profibus master's Engineering Tool - Check the configuration being used - it is possible that the number of configured I/O modules is too high
Off	Blinking	SERVICE MODE - BOOT The bus controllers node number has been set to 255 (\$FF) - after 2 seconds the bus controller starts in Service mode	Set a valid node number
Single flash	Single flash	HARDWARE FAULT	

Table 35: BC0063 - status LED description

4.4.11 Further documentation and import files (GSD)

Further documentation about Profibus DP bus controller functions, as well as the necessary import files for the master engineering tool, are available for download from the B&R homepage (www.br-automation.com).

4.5 BC0073

4.5.1 General information

The bus controller BC0073 lets you link X2X connect I/O nodes to CAN I/O. CAN I/O is a transfer protocol based on the CAN bus standard and is fully integrated into the B&R system. From the user's point of view, it doesn't matter if I/O points are operated locally or remotely via CAN I/O.

- Fieldbus: CAN bus
- Automatic firmware update via the fieldbus
- Integrated I/O access in B&R Automation Studio™
- Integrated terminating resistance

4.5.2 Order data

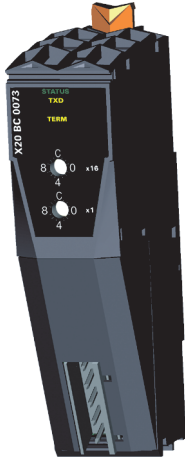
Model number	Short description	Image
	Bus controller	
X20BC0073	X20 bus controller fieldbus interface, 1 CAN I/O interface, status indicator LEDs, order 1 x TB2105 terminal block separately	
	Required accessories	
0TB2105.9010	Accessory, terminal block, 5-pin, screw clamps, 2.5 mm²	
0TB2105.9110	Accessory, terminal block, 5-pin, cage clamps, 2.5 mm²	
X20TB12	Standard X20 terminal block (12-pin)	
X20PS9400	X20 supply module for bus controller and internal I/O supply, X2X link bus supply	
X20BB80	Basic X20 bus controller, for bus controller fieldbus interface and bus controller supply module, X20 connection	

Table 36: BC0073 - order data

4.5.3 Technical data

Product ID	BC0073
Short description	
Bus controller	CAN I/O slave
Fieldbus	
Type	CAN I/O slave
Type	5-pin multipoint connector
Maximum distance	1,000 m
Maximum transfer rate	1 MBit/s
Determination of transfer rate	Automatic transfer rate detection, permanently set or from internal EEPROM
General information	
Status indicators	Module status, bus function, data transfer, terminating resistance
Diagnostics Module status Bus function Data transfer Terminating resistors	Yes, with status LED and software status Yes, with status LED Yes, with status LED Yes, with status LED
Electrical isolation Fieldbus - X2X bus Fieldbus - I/O	No Yes
Power consumption - bus	1.5 W
Certification	CE, C-UL-US (in development), GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	37.5 ±0.2 mm
Note	Order standard terminal block 1 x TB2105 separately Order standard terminal block 1 x X20TB12 separately Order supply module 1 x X20PS9400 separately Order bus controller base 1 x X20BB80 separately

Table 37: BC0073 - technical data

4.5.4 Additional technical data

Product ID	BC0073
Fieldbus	
Controller	SJA 1000 controller
Processor	32-bit RISC processor
Memory	512 KB
Terminating resistance	Integrated in the module
General information	
B&R ID code	\$1F1D

Table 38: BC0073 - additional technical data

4.5.5 Status LEDs

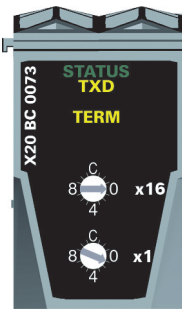
Image	LED	Color	Status	Description
	STATUS ¹⁾	Green	Off	No supply via CAN fieldbus
			Blinking	Preoperational mode
			On	RUN mode
		Red	On	CAN connection reports BusOff status
		Blinking green / red		Transfer rate detection in progress
		Blinking green / single red flash		Preoperational mode / CAN connection reports that the warning limit has been reached
		Steady green / single red flash		RUN mode / CAN connection reports that the warning limit has been reached
	TXD	Yellow	Off	The bus controller is not transmitting any data via the CAN I/O fieldbus
			On	The bus controller is transmitting data via CAN I/O fieldbus
	TERM	Yellow	Off	The terminating resistance integrated in the bus controller is turned off
			On	The terminating resistance integrated in the bus controller is turned on

Table 39: BC0073 - status indicators

1) The STATUS LED is a green/red dual LED.

4.5.6 Operating and connection elements

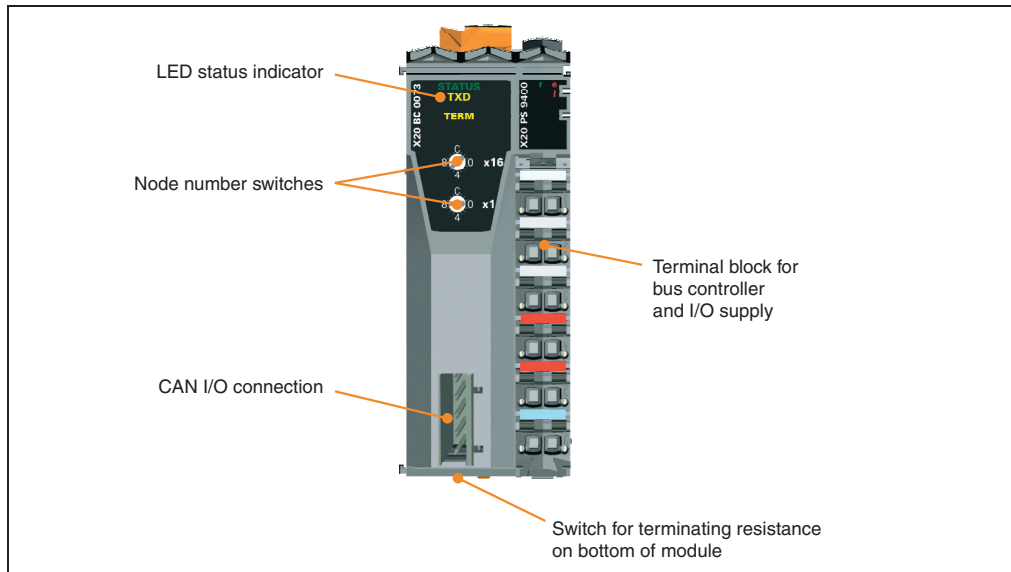


Figure 33: BC0073 - operational and connection elements

4.5.7 CAN interface

The CAN interface is a 5-pin multipoint connector. The TB2105 terminal block must be ordered separately.

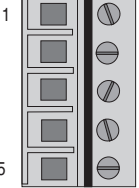
5-pin terminal block	Terminal	Assignment	
 TB2105	1	CAN_L	CAN ground
	2	CAN_L	CAN low
	3	SHLD	Shield
	4	CAN_H	CAN high
	5	Reserved	

Table 40: BC0073 - pin assignments - CAN interface

4.5.8 Terminating resistors

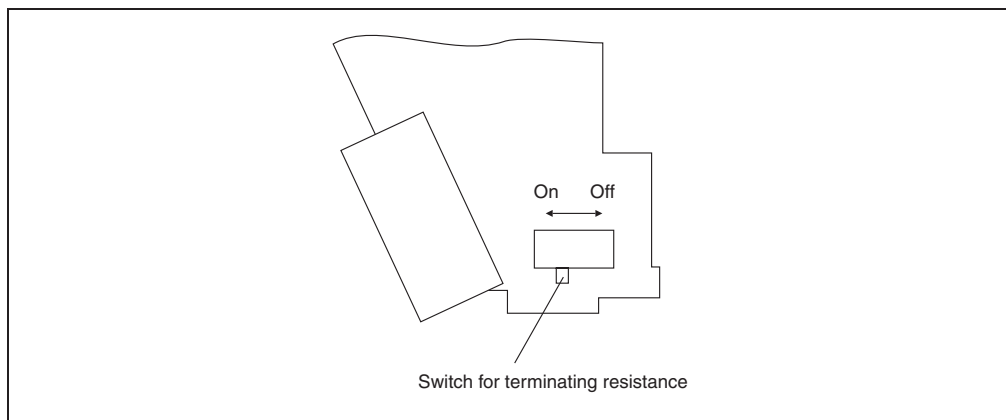


Figure 34: BC0073 - terminating resistance

A terminating resistance is already integrated on the bus controller BC0073. It is turned on and off with a switch on the bottom of the housing. An active terminating resistance is indicated by the TERM LED.

4.5.9 Node numbers and transfer rate

Node numbers and transfer rates are configured using both bus controller number switches. The switch positions \$00 - \$40 and \$60 activate the automatic transfer rate detection (see section 4.5.10 "Automatic transfer rate detection" on page 111). The rest of the switch positions have a fixed transfer rate (see table).

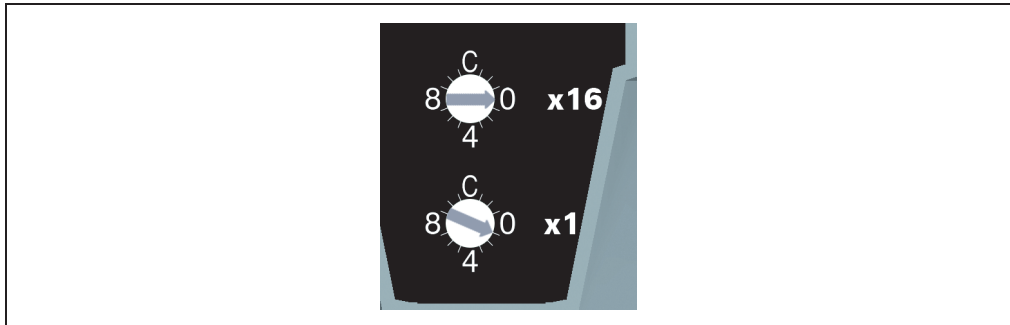


Figure 35: BC0073 - number switches for node numbers and transfer rates

Switch position	Node number	Transfer rate
\$00 ¹⁾	From EEPROM	From EEPROM
\$01 - \$3F	1 - 63	Automatic
\$40 ¹⁾	From EEPROM	From EEPROM
\$41 - \$5F	1 - 31	1000 kBit/s
\$60 ¹⁾	From EEPROM	From EEPROM
\$61 - \$7F	1 - 31	800 kBit/s
\$80	Reserved	-
\$81 - \$9F	1 - 31	500 kBit/s
\$A0	Reserved	-
\$A1 - \$BF	1 - 31	250 kBit/s
\$C0	Reserved	-
\$C1 - \$DF	1 - 31	125 kBit/s
\$E0	Reserved	-
\$E1 - \$FE	1 - 31	20 kBit/s
\$FF	Reserved	-

Table 41: BC0073 - node numbers and transfer rates

- 1) When one of these numbers is configured, the bus controller uses the operating parameters from the internal EEPROM. The EEPROM is programmed using the CANIO library.

4.5.10 Automatic transfer rate detection

After booting, the bus controller BC0073 goes into "Listen Only" mode. That means the bus controller behaves passively on the bus and only listens.

The BC0073 tries to receive valid objects. If errors occur during receiving, the controller switches to the next transfer rate from the search table.

If no objects are received, all transfer rates are tested cyclically. This procedure is repeated until valid objects are received.

Starting transfer rate

The bus controller begins the search with this transfer rate. The starting transfer rate can be defined in two different ways:

- Read from EEPROM
- The last recognized transfer rate is used to begin the search after a software reset (command code 20)

Search table

The controller tests the transfer rate according to this table. Beginning with the starting transfer rate, the controller switches to the next lower transfer rate. At the end of the table, the bus controller starts searching from the beginning again.

Transfer rate
1000 kBit/s
500 kBit/s
250 kBit/s
125 kBit/s
50 kBit/s
20 kBit/s
10 kBit/s

Table 42: BC0073 - transfer rate search table

4.6 BC0083

4.6.1 General information

The BC0083 bus controller makes it possible to connect X2X Link I/O nodes to ETHERNET Powerlink V1 and V2. It is also possible to operate the X2X Link cycle synchronously 1:1 or asynchronous to Powerlink using a prescaler.

ETHERNET Powerlink is a standard protocol for Fast Ethernet with true real-time properties. The ETHERNET Powerlink Standardization Group (EPSG, www.ethernet-powerlink.org) ensures that the standard remains open and is continually developed.

- ETHERNET Powerlink V1 and V2
- I/O configuration and firmware update via the fieldbus
- Integrated hub for efficient cabling

4.6.2 Order data

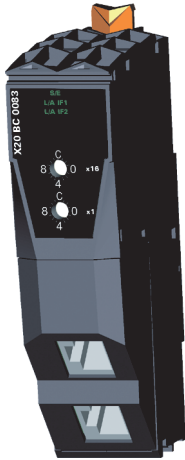
Model number	Short description	Image
	Bus controller	
X20BC0083	X20 bus controller fieldbus interface, ETHERNET Powerlink interface, integrated 2x hub, status indicator LEDs, 2 x RJ45 connection	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20PS9400	X20 supply module for bus controller and internal I/O supply, X2X link bus supply	
X20BB80	Basic X20 bus controller, for bus controller fieldbus interface and bus controller supply module, X20 connection	

Table 43: BC0083 - order data

4.6.3 Technical data

Product ID	BC0083
Short description	
Bus controller	ETHERNET Powerlink Controlled Node (EPL CN)
Fieldbus	
Type	ETHERNET Powerlink 100 Base-T (ANSI/IEEE 802.3)
Design	Internal 2x hub, 2 x shielded RJ45 port
Cable length	Max. 100 m between two stations (segment length)
Transfer rate	100 Mbit/s
General information	
Status indicators	Module status, bus function
Diagnostics Module status Bus function	Yes, with status LED and software status Yes, with status LED and software status
Electrical isolation Fieldbus - X2X bus Fieldbus - I/O	Yes Yes
Power consumption - bus	2.0 W
Certification	CE, C-UL-US (in development), GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	37.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order supply module 1 x X20PS9400 separately Order bus controller base 1 x X20BB80 separately

Table 44: BC0083 - technical data

4.6.4 Additional technical data

Product ID	BC0083
Fieldbus	
Controller	EPL MAC
Processor	32-bit RISC processor
Memory	512 KB
General information	
B&R ID code	\$1F1E

Table 45: BC0083 - additional technical data

4.6.5 Status LEDs

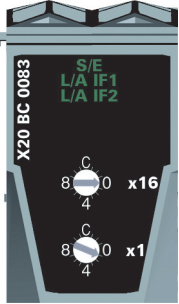
Image	LED	Color	Status	Description
	S/E ¹⁾	Green/red		Status/Error LED. The status of the LED is described in section 4.6.6 "S/E LED" on page 115.
	L/A IFx	Green	On	A link to the remote station has been established.
			Blinking	A link to the remote station has been established. The LED blinks when Ethernet activity is present on the bus.

Table 46: BC0083 - status indicators

1) The Status/Error LED is a green/red dual LED.

4.6.6 S/E LED

The Status/Error LED is a green/red dual LED. The color green (status) is superimposed on the color red (error).

Color red - error	Description
On	The bus controller has encountered an error (failed Ethernet frames, burst of collisions on the network, etc.). Note: The LED blinks red several times immediately after startup. This is not an error.

Table 47: BC0083 - Status/Error LED lit red: LED indicates error

Green - status	Description
Off	The bus controller is either not getting power, or it is NOT_ACTIVE. The bus controller waits in this state for about 5 seconds after a restart. No communication with the bus controller is possible. If no ETHERNET Powerlink communication is detected during these 5 s, the bus controller goes into the BASIC_ETHERNET state (blinking). If, however, ETHERNET Powerlink communication is detected during this time, the bus controller goes directly into the PRE_OPERATIONAL_1 status (single flash).
Blinking green (approx. 10 Hz)	The bus controller did not detect any ETHERNET Powerlink communication. In this state you can communicate directly with the bus controller using UDP. If an ETHERNET Powerlink communication is detected while in this state, the bus controller goes into the PRE_OPERATIONAL_1 state (single flash).
Single flash (approx. 1 Hz)	The bus controller is in PRE_OPERATIONAL_1. With operation on an ETHERNET Powerlink V1 master, the bus controller goes directly into PRE_OPERATIONAL_2. With operation on an ETHERNET Powerlink V2 manager, the CN (Controlled Node) waits for the reception of a SoC frame and then switches over to PRE_OPERATIONAL_2 (double flash).
Double flash (approx. 1 Hz)	The bus controller is in PRE_OPERATIONAL_2. In this state the bus controller is normally configured by the manager. A command (ETHERNET Powerlink V2) or setting the data valid flag in the output data (ETHERNET Powerlink V1) then switches the state to READY_TO_OPERATE (triple flash).
Triple flash (approx. 1 Hz)	The bus controller is in READY_TO_OPERATE. In an ETHERNET Powerlink V2 network, the manager then switches via command to OPERATIONAL. In an ETHERNET Powerlink V1 network, the bus controller then switches automatically to OPERATIONAL as soon as input data are present.
On	The bus controller is in OPERATIONAL.
Blinking (approx. 2.5 Hz)	The bus controller is in STOPPED. No output data are produced and no input data are received. Only the appropriate command from the manager can enter or leave this state.

Table 48: BC0083 - Status/Error LED lit green: LED indicates operation

4.6.7 Operating and connection elements

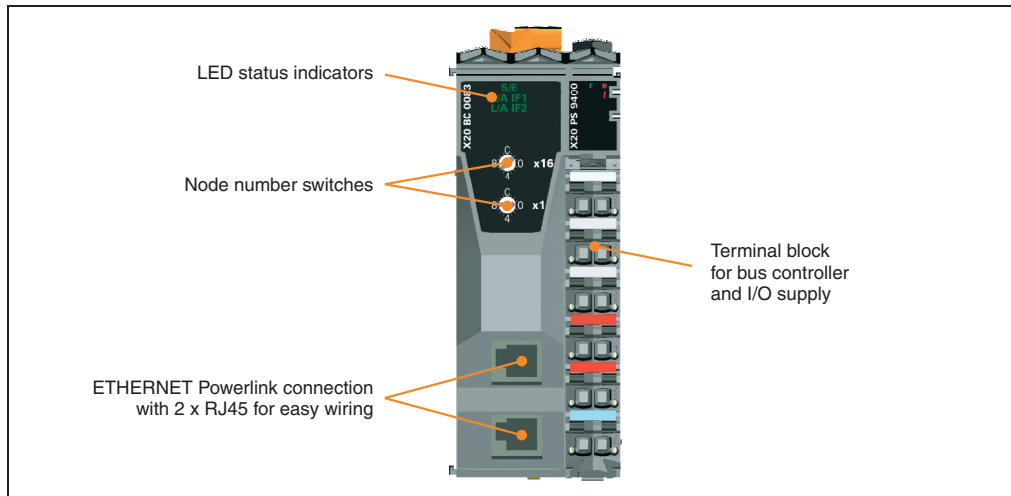


Figure 36: BC0083 - operational and connection elements

4.6.8 ETHERNET Powerlink station number

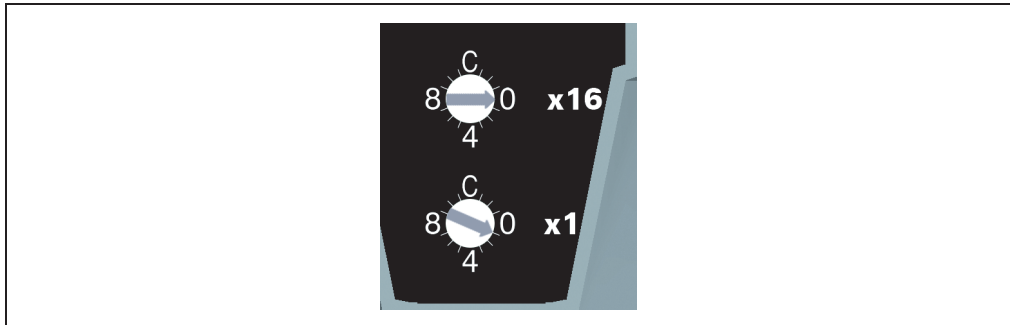


Figure 37: BC0083 - station number switches

The station number for the Powerlink station is set using both number switches. Station numbers are permitted between \$01 and \$FD.

Switch position	Description
\$00	Reserved for manager station, switch position is not permitted.
\$01 - \$FD	Station number for Powerlink station.
\$FE	Reserved, switch position is not permitted.
\$FF	Reserved, switch position is not permitted.

Table 49: BC0083 - station numbers

4.6.9 RJ45 ports

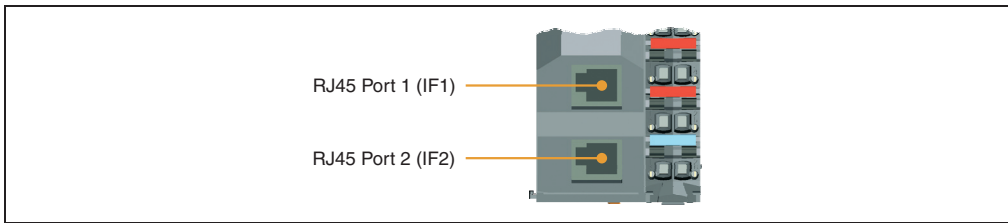


Figure 38: BC0083 - RJ45 ports

Pin	Assignment
1	RXD
2	RXD\
3	TXD
4	Termination
5	Termination
6	TXD\
7	Termination
8	Termination

Table 50: BC0083 - pin assignments - RJ45 port

RXD ... Receive Data

TXD ... Transmit Data

4.6.10 SG3

The BC0083 module is currently not supported on SG3 targets.

4.6.11 SG4

The firmware update takes place automatically. The firmware is a component of the PLC operating system of B&R Automation Runtime™ (with AR version V2.85 or higher).

5. Bus controller system modules

5.1 General information

The X20 System bus controllers are made up of a bus controller fieldbus interface, a bus controller system module, and a 12-pin terminal block.

Bus controller system modules include the base module BB80 and the supply module PS9400.

5.1.1 Overview

Model number	Short description
X20BB80	Basic X20 bus controller, for bus controller fieldbus interface and bus controller supply module, X20 connection
X20PS9400	X20 supply module for bus controller and internal I/O supply, X2X link bus supply

Table 51: Overview of bus controller system modules

5.2 BB80

5.2.1 General information

The BB80 bus module is the basis for all X20 bus controllers. The left and right locking plates are included in the delivery (for info on installation see chapter 6 "Mechanical design", page 429).

- Basis for all X20 bus controllers

5.2.2 Order data

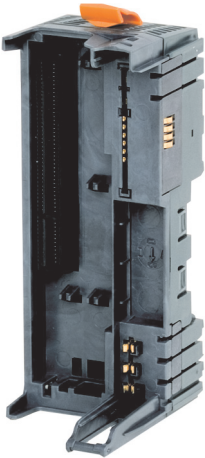
Model number	Short description	Image
	Bus module	
X20BB80	Basic X20 bus controller, for bus controller fieldbus interface and bus controller supply module, X20 connection	

Table 52: BB80 - order data

5.2.3 Technical data

Product ID	BB80
Short description	
Bus module	Bus controller base - backplane for bus controller fieldbus interface and bus controller supply module
General information	
Power consumption Bus I/O internal	- -
Certification	CE, C-UL-US (in development), GOST-R (in development)

Table 53: BB80 - technical data

Product ID	BB80
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	37.5 ±0.2 mm

Table 53: BB80 - technical data (cont.)

5.2.4 Potential control

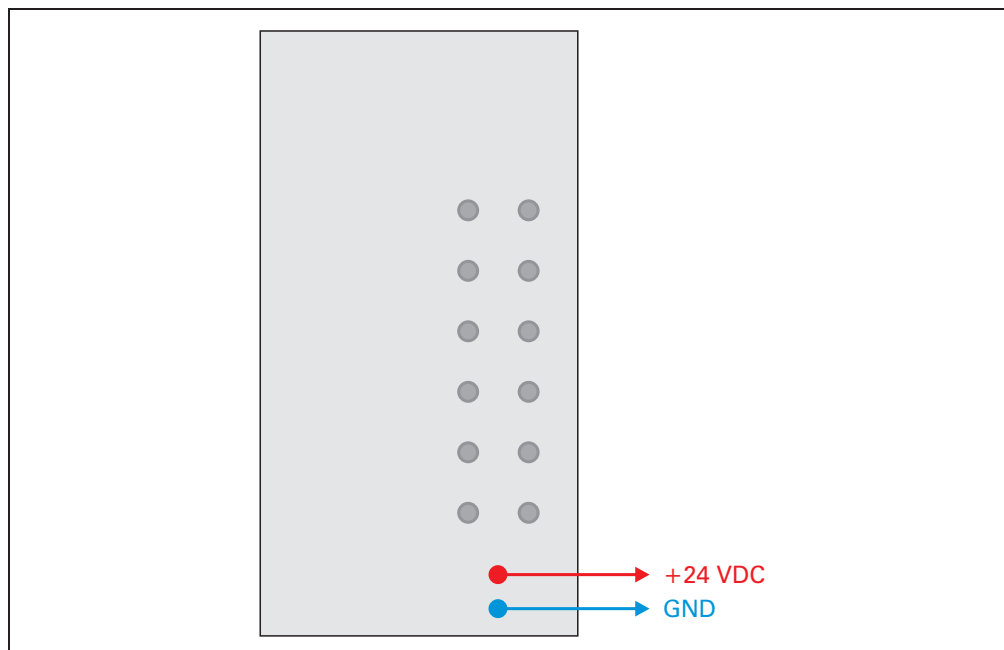


Figure 39: BB80 - potential control